Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 2

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 3

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.961506
<D> / site : 0.493584

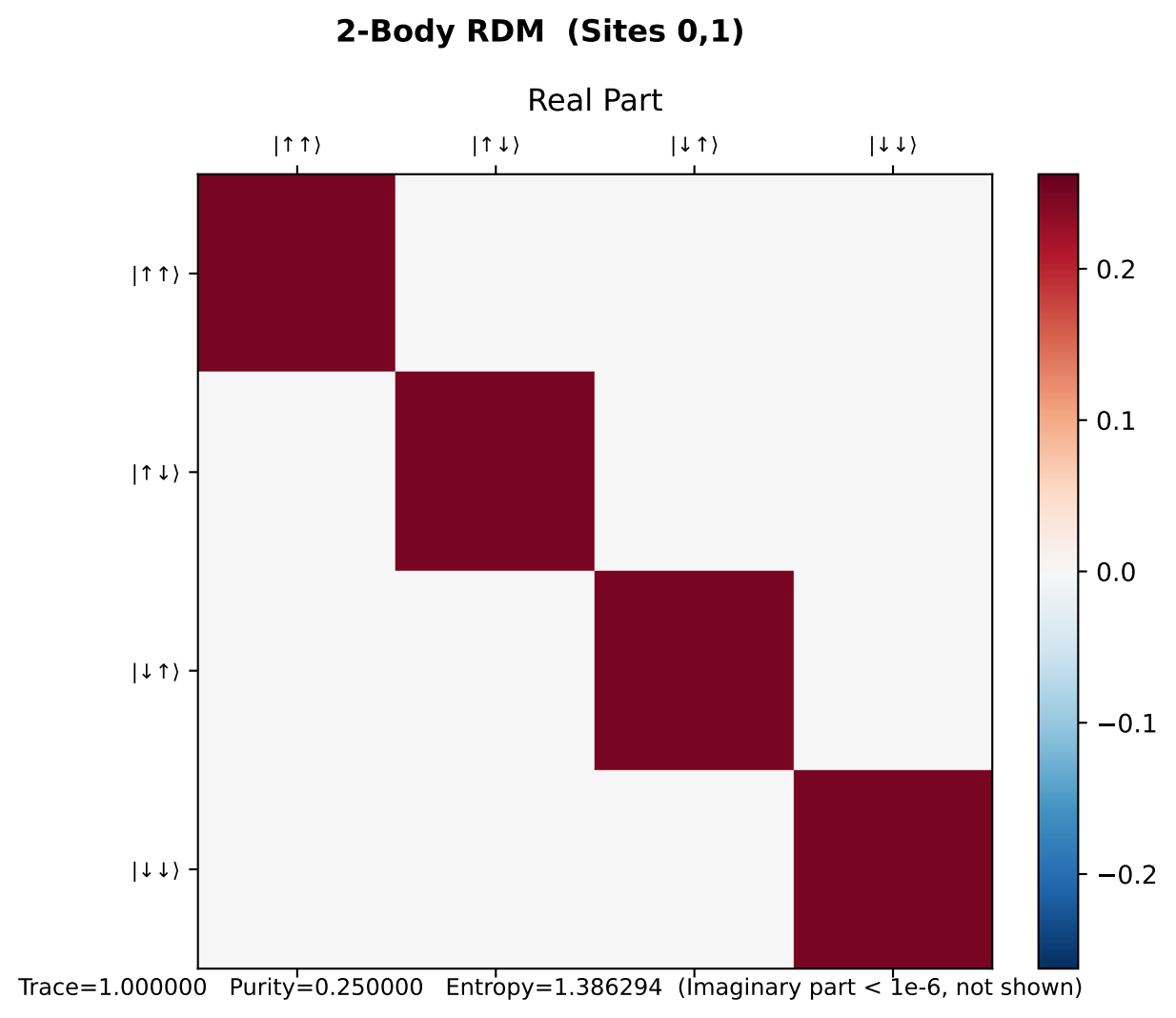
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Body order : 2
Clusters   : 15

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Summary — 2-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1	0.250000	—	—	—	—	—	MIXED
0,2	0.250000	—	—	—	—	—	MIXED
0,3	0.250000	—	—	—	—	—	MIXED
0,4	0.250000	—	—	—	—	—	MIXED
0,5	0.250123	—	—	—	—	—	MIXED
1,2	0.250000	—	—	—	—	—	MIXED
1,3	0.297722	—	—	—	—	—	MIXED
1,4	0.251106	—	—	—	—	—	MIXED
1,5	0.250000	—	—	—	—	—	MIXED
2,3	0.251113	—	—	—	—	—	MIXED
2,4	0.297595	—	—	—	—	—	MIXED
2,5	0.250000	—	—	—	—	—	MIXED
3,4	0.251896	—	—	—	—	—	MIXED
3,5	0.250004	—	—	—	—	—	MIXED
4,5	0.250005	—	—	—	—	—	MIXED


2-Body RDM (Sites 0,1)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,1)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500000e-01
Imaginary part	: < 1e-6 (not shown)

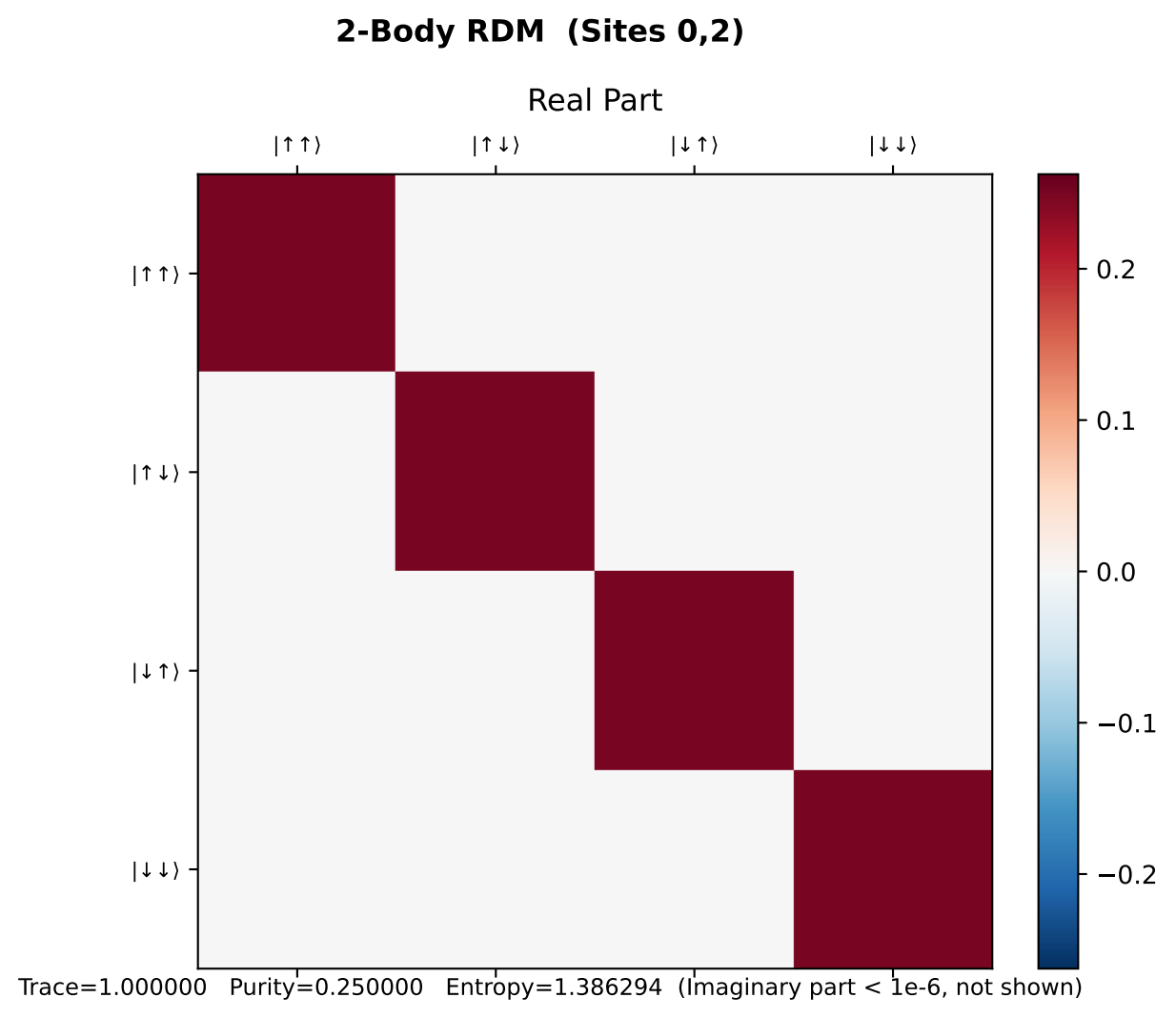
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,2)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,2)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500000e-01
Imaginary part	: < 1e-6 (not shown)

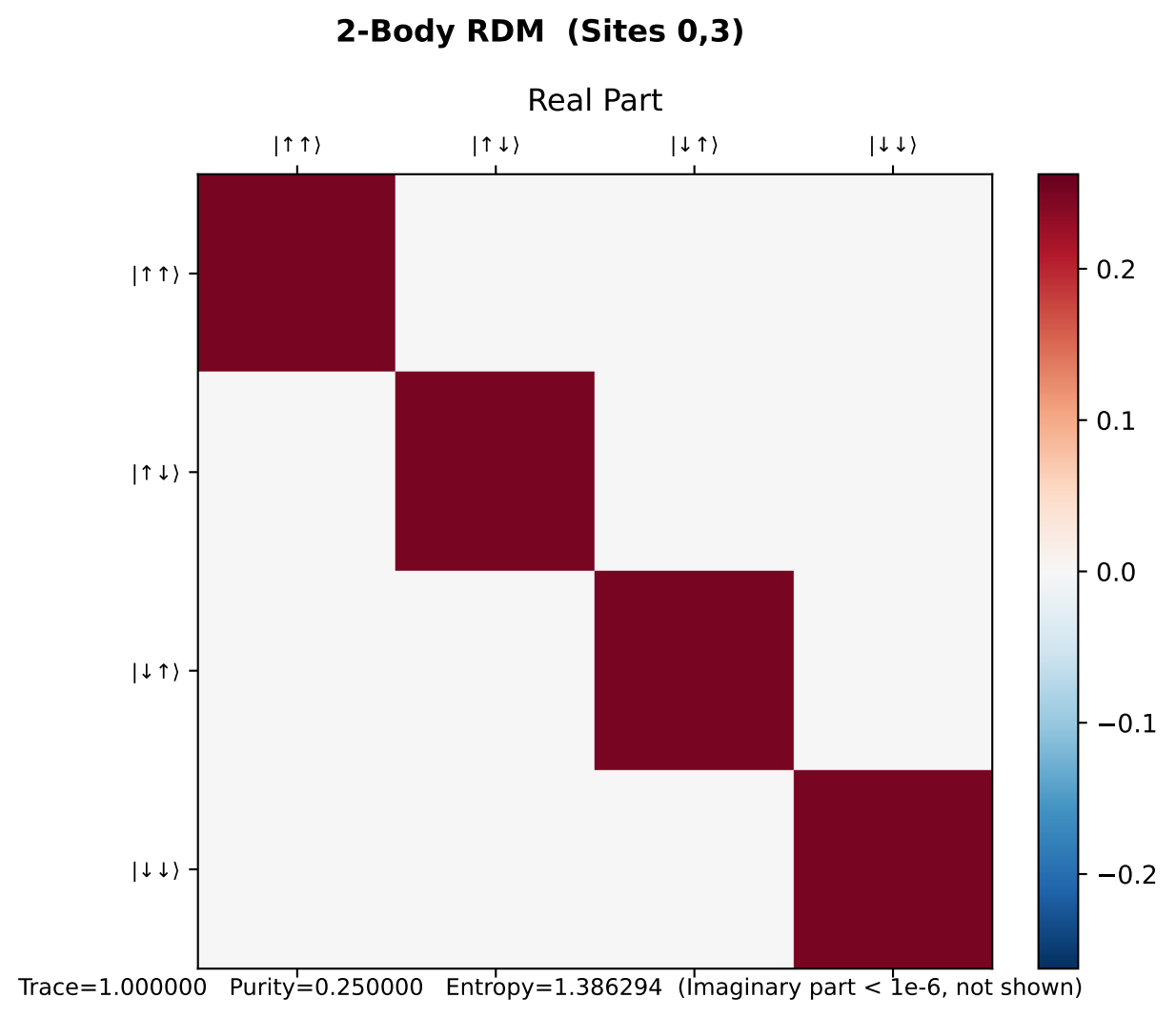
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,3)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500001e-01
Imaginary part	: < 1e-6 (not shown)

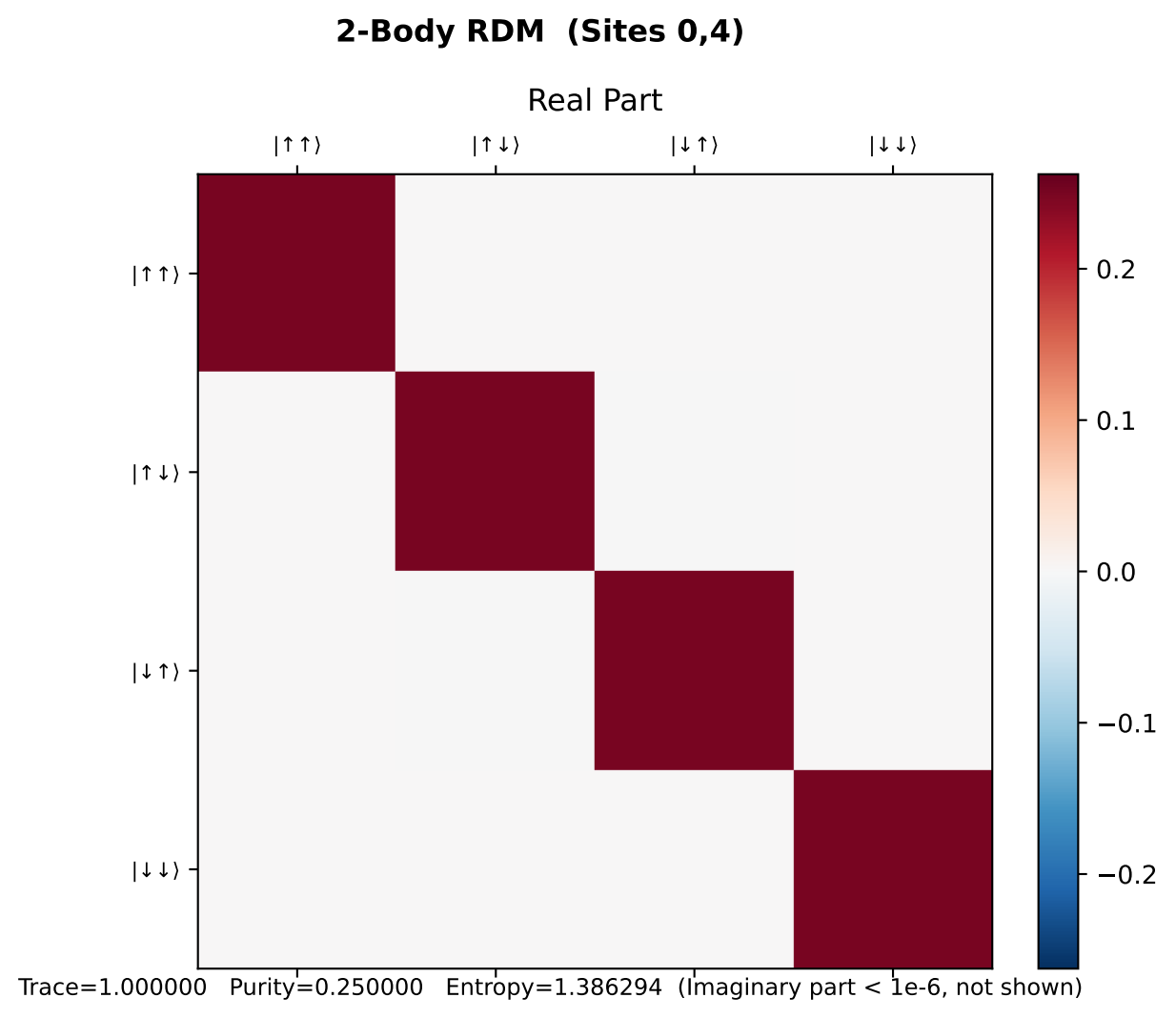
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500001e-01
Imaginary part	: < 1e-6 (not shown)

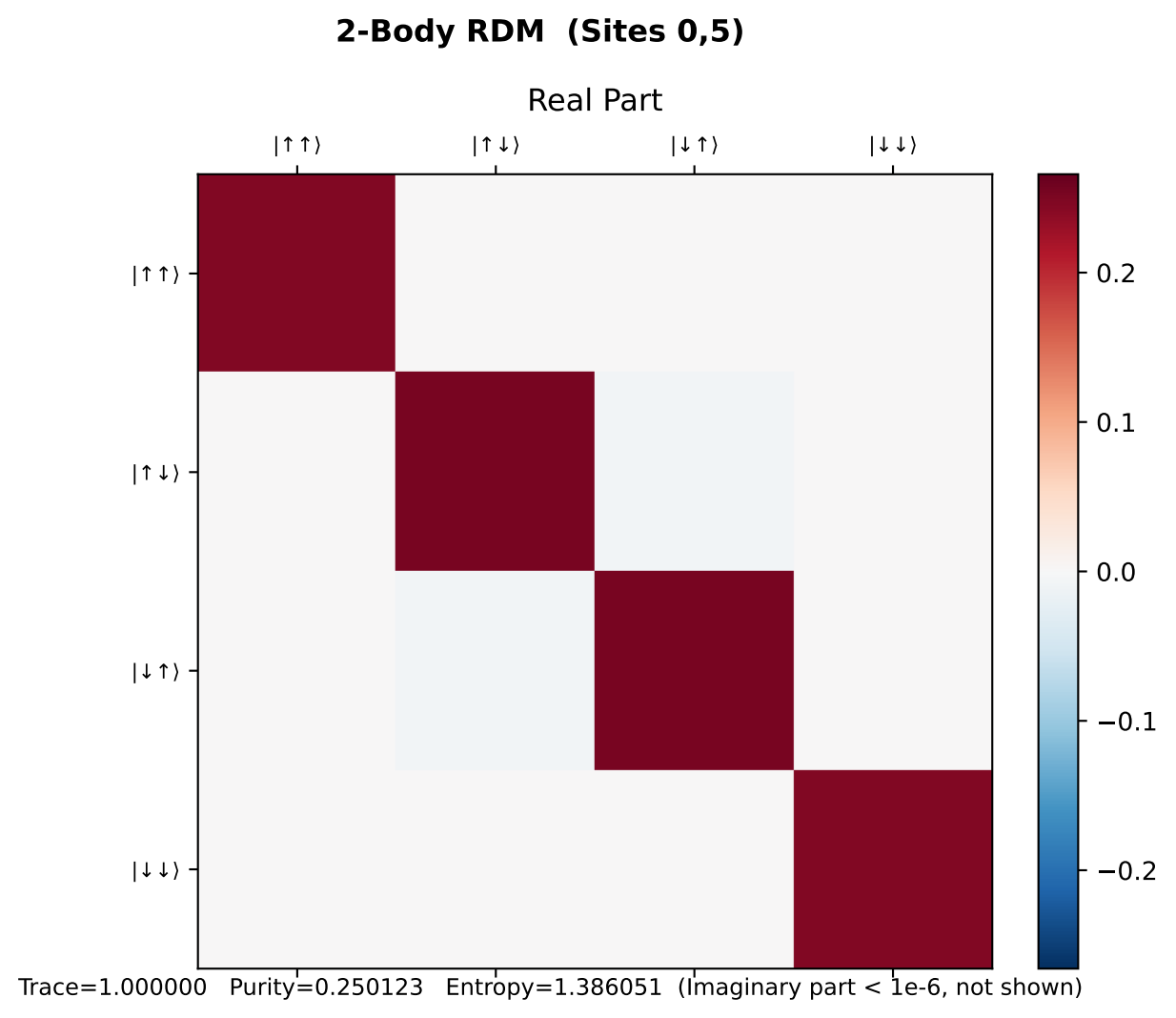
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,5)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250123 Entropy=1.386051 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2501228452
Von Neumann entropy	: 1.3860507211
Eigenvalue min	: 2.468005e-01
Eigenvalue max	: 2.595986e-01
Imaginary part	: < 1e-6 (not shown)

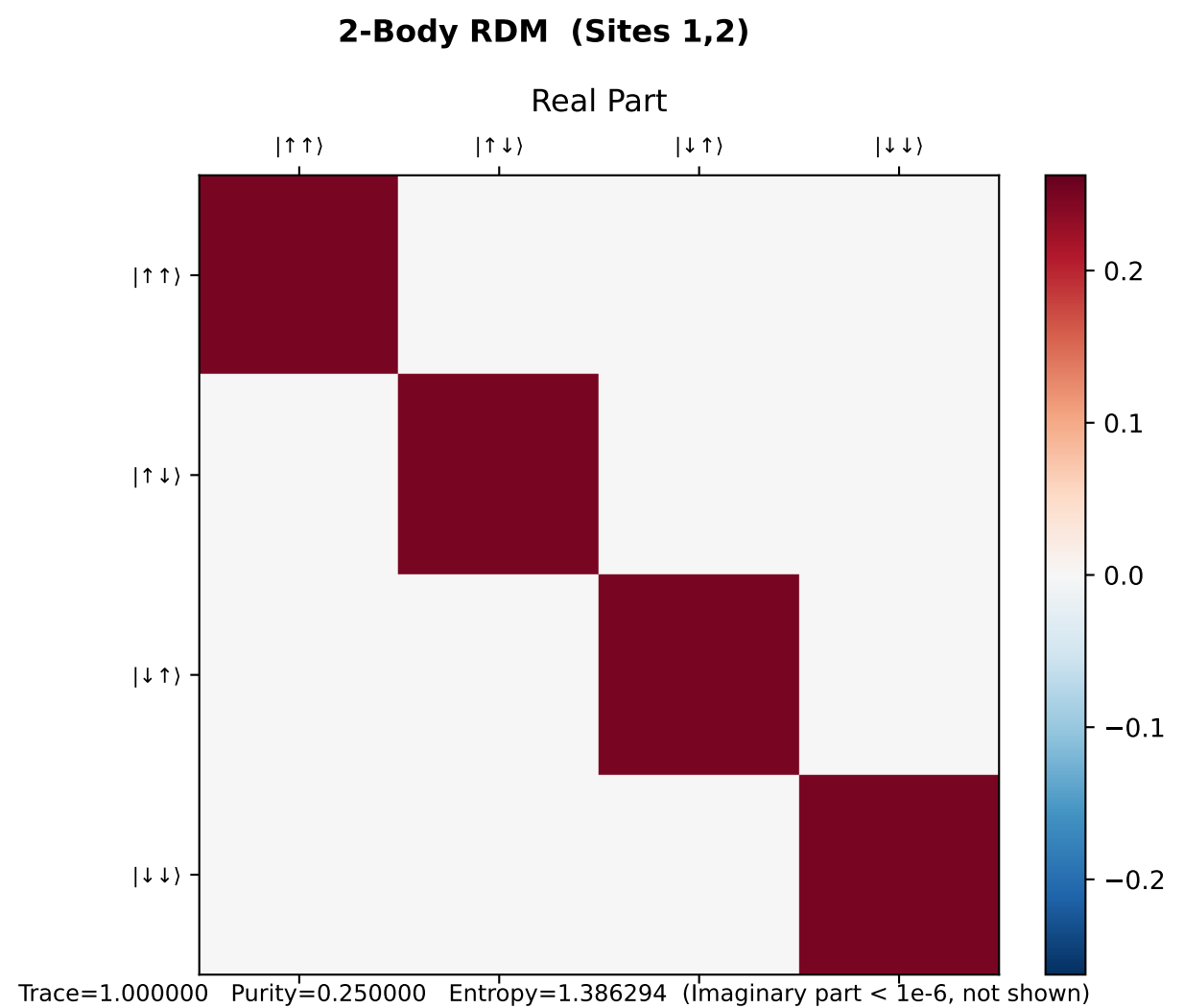
GME Analysis
==========

gamma = Tr(rho^2) = 0.25012285

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 1,2)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 1,2)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000695
Von Neumann entropy	: 1.3862942220
Eigenvalue min	: 2.497716e-01
Eigenvalue max	: 2.500761e-01
Imaginary part	: < 1e-6 (not shown)

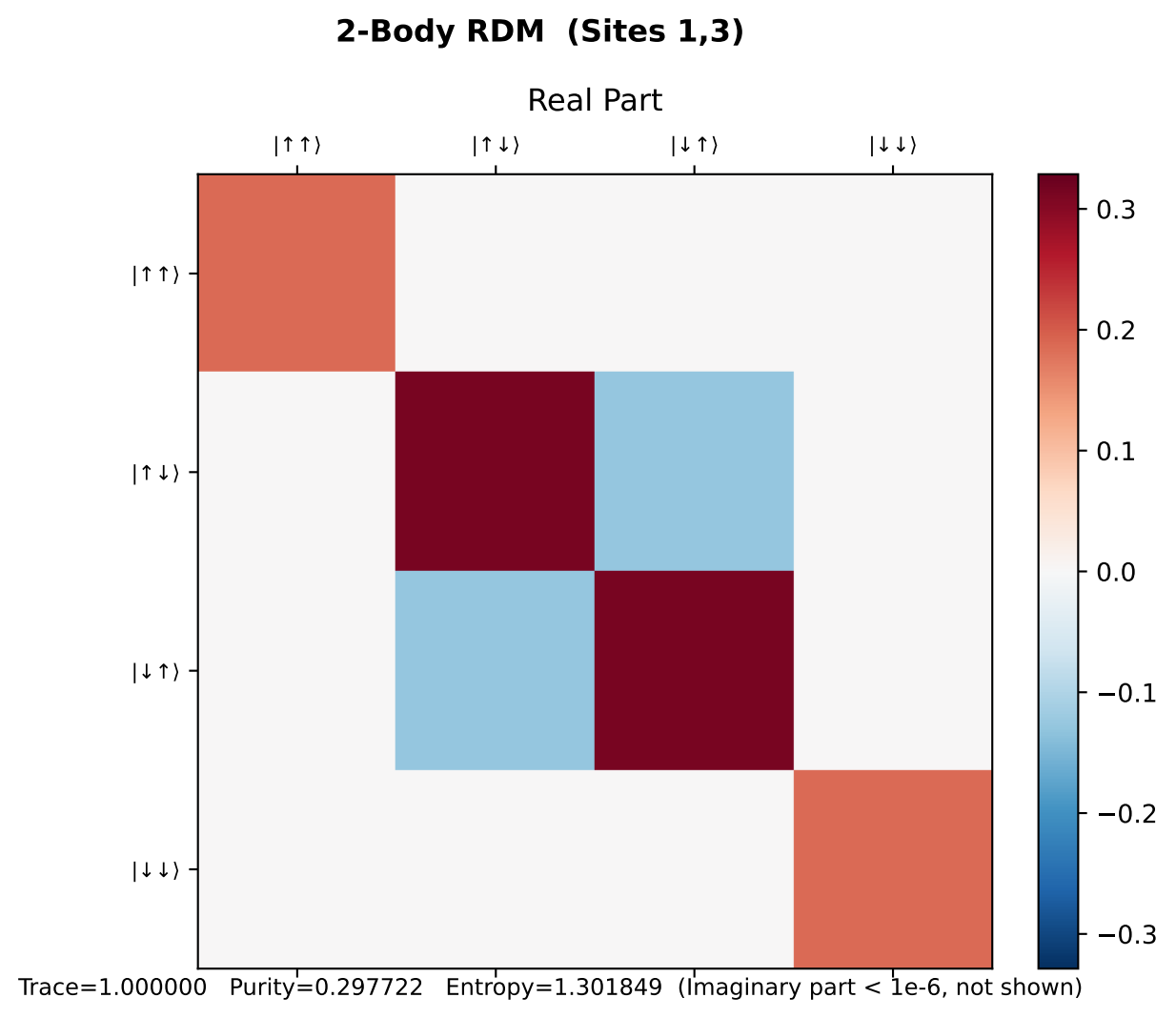
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000007

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 1,3)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=0.297722 Entropy=1.301849 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 1,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2977220375
Von Neumann entropy	: 1.3018488829
Eigenvalue min	: 1.869378e-01
Eigenvalue max	: 4.391865e-01
Imaginary part	: < 1e-6 (not shown)

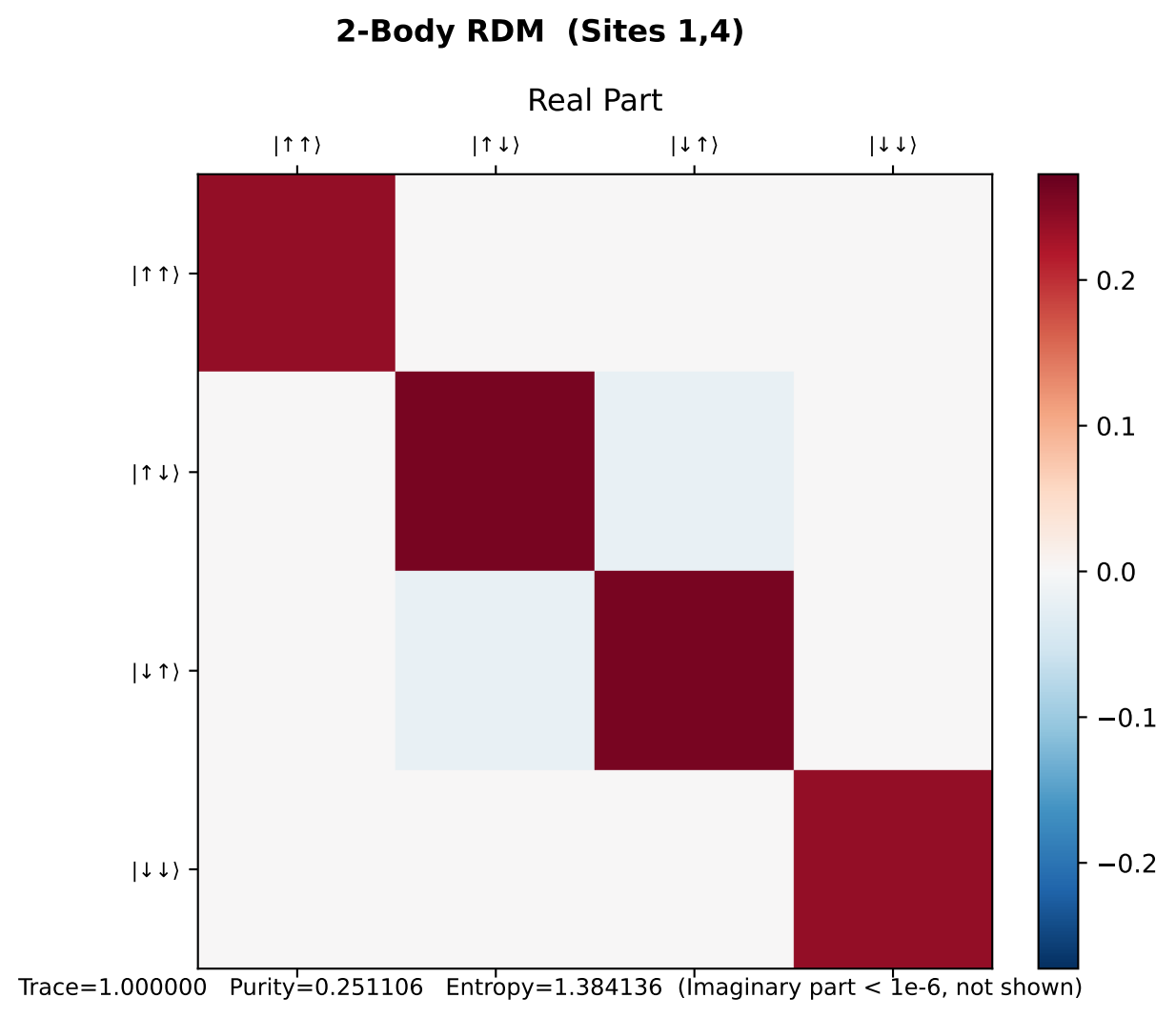
GME Analysis
==========

gamma = Tr(rho^2) = 0.29772204

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 1,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.251106 Entropy=1.384136 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 1,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2511058423
Von Neumann entropy	: 1.3841357214
Eigenvalue min	: 2.404003e-01
Eigenvalue max	: 2.787990e-01
Imaginary part	: < 1e-6 (not shown)

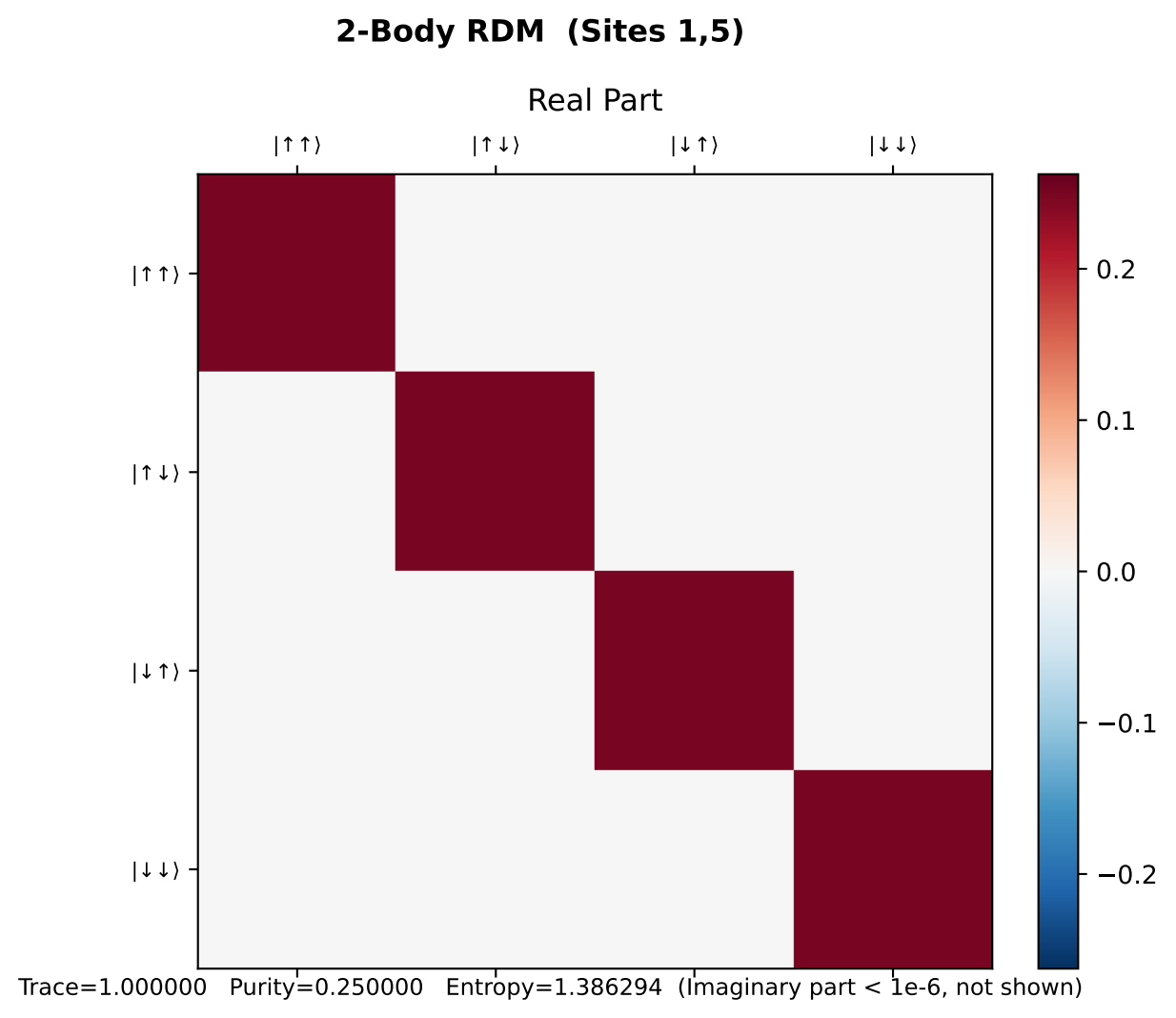
GME Analysis
==========

gamma = Tr(rho^2) = 0.25110584

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 1,5)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 1,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000085
Von Neumann entropy	: 1.3862943441
Eigenvalue min	: 2.499734e-01
Eigenvalue max	: 2.500798e-01
Imaginary part	: < 1e-6 (not shown)

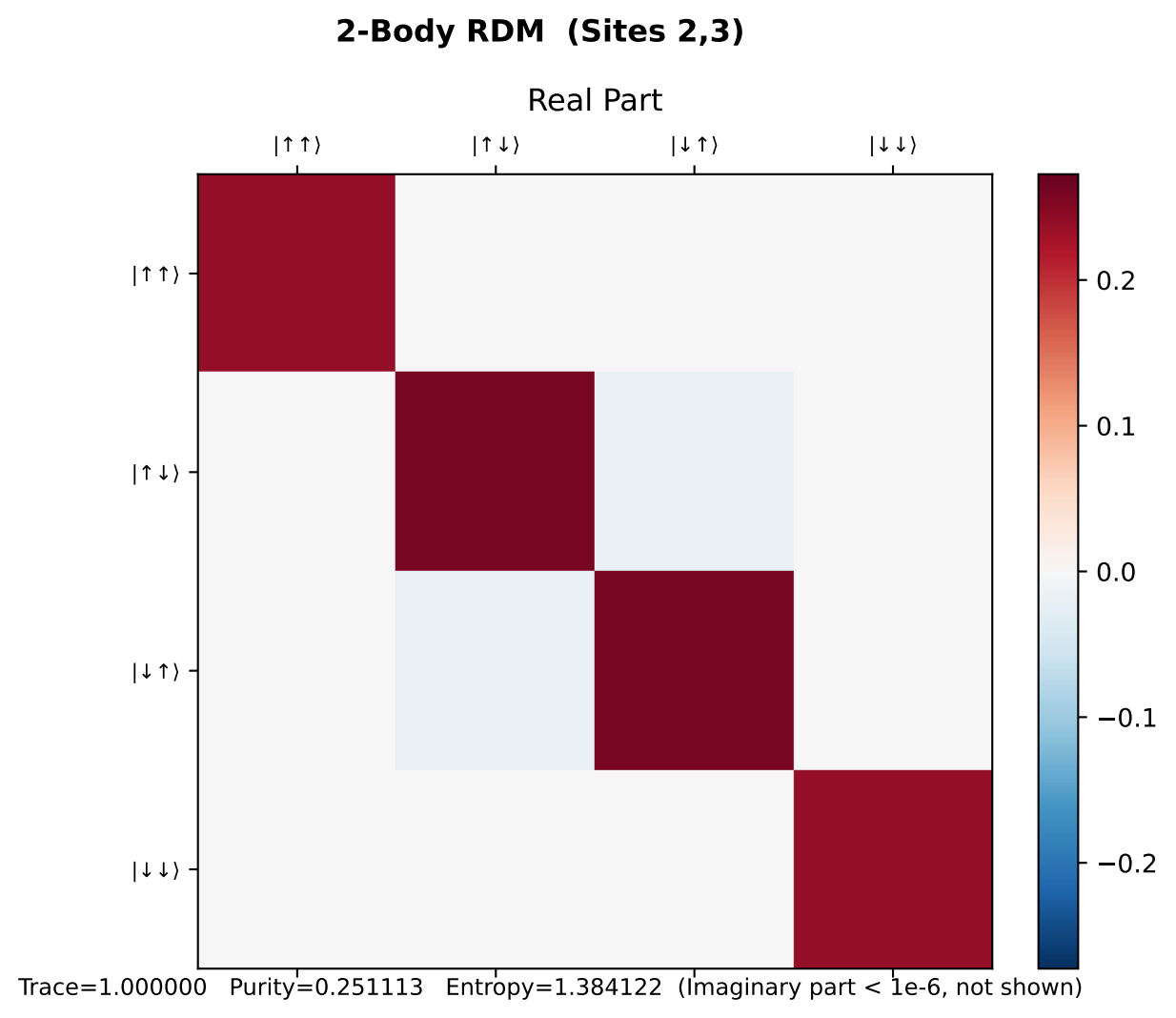
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000001

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 2,3)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.251113 Entropy=1.384122 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 2,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2511131854
Von Neumann entropy	: 1.3841215533
Eigenvalue min	: 2.403685e-01
Eigenvalue max	: 2.788944e-01
Imaginary part	: < 1e-6 (not shown)

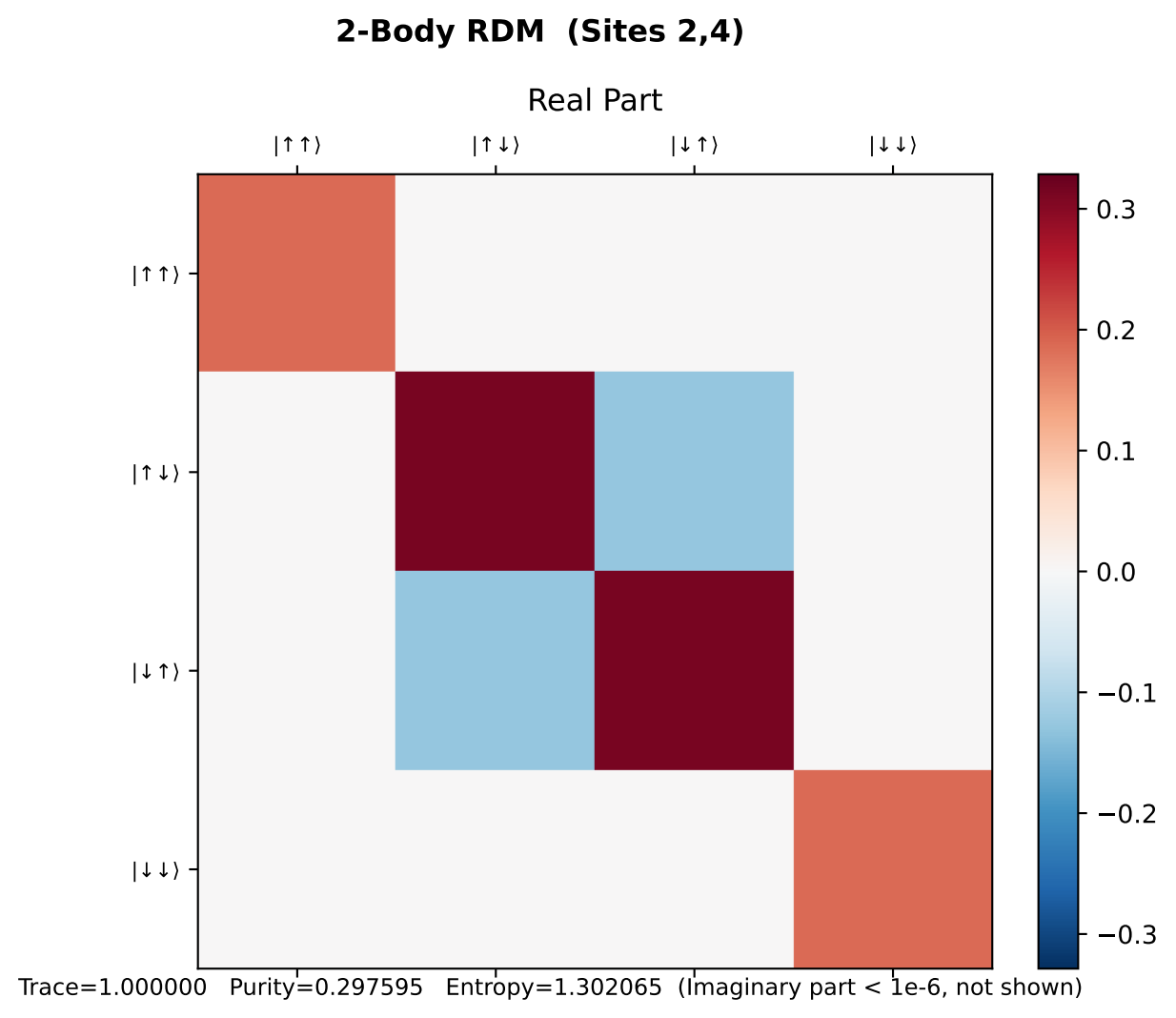
GME Analysis
==========

gamma = Tr(rho^2) = 0.25111319

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 2,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=0.297595 Entropy=1.302065 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 2,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2975946030
Von Neumann entropy	: 1.3020646526
Eigenvalue min	: 1.870221e-01
Eigenvalue max	: 4.389337e-01
Imaginary part	: < 1e-6 (not shown)

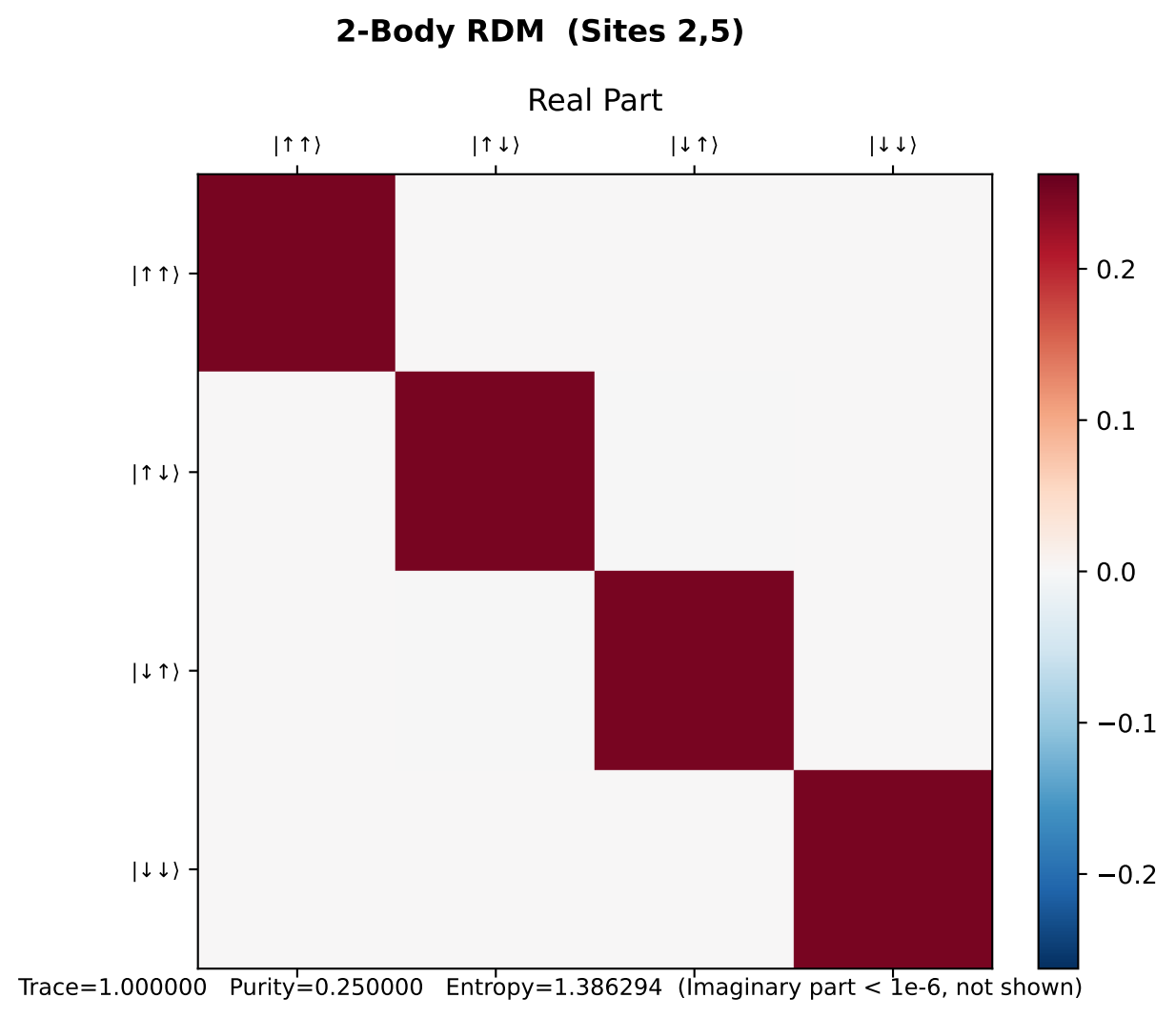
GME Analysis
==========

gamma = Tr(rho^2) = 0.29759460

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 2,5)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 2,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000112
Von Neumann entropy	: 1.3862943387
Eigenvalue min	: 2.499694e-01
Eigenvalue max	: 2.500918e-01
Imaginary part	: < 1e-6 (not shown)

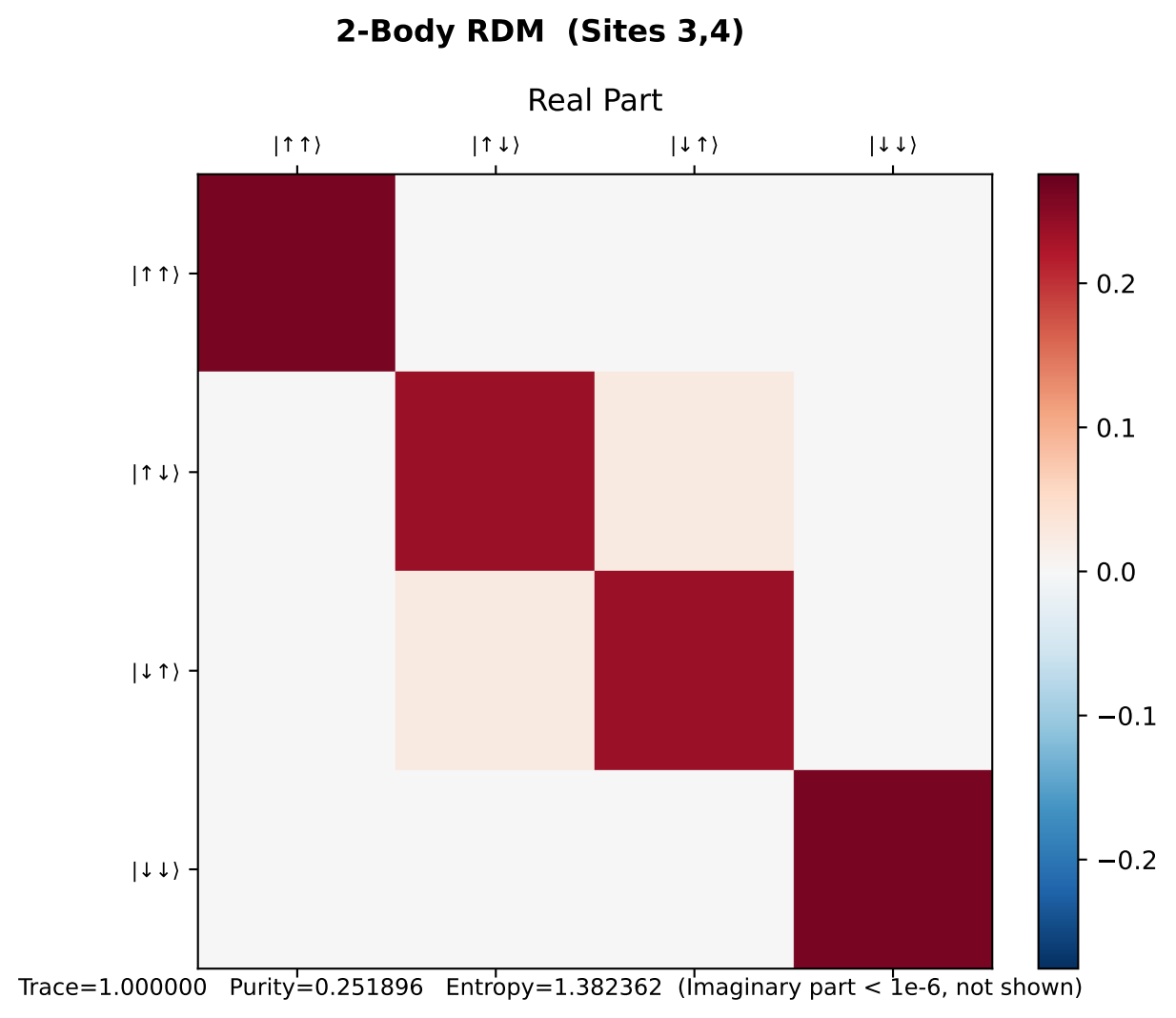
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000001

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 3,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.251896 Entropy=1.382362 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 3,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2518964177
Von Neumann entropy	: 1.3823621140
Eigenvalue min	: 2.122864e-01
Eigenvalue max	: 2.625712e-01
Imaginary part	: < 1e-6 (not shown)

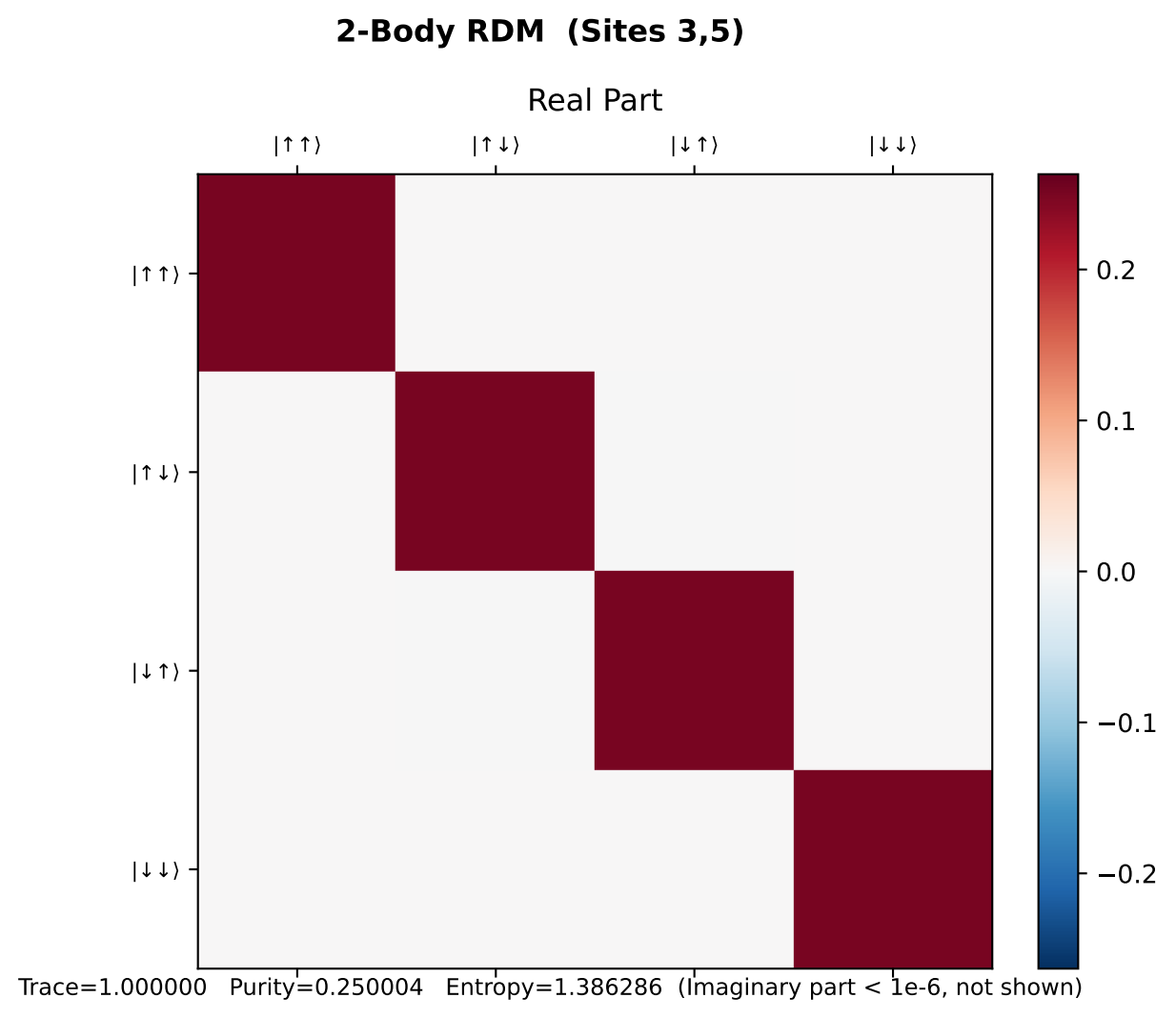
GME Analysis
==========

gamma = Tr(rho^2) = 0.25189642

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 3,5)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250004 Entropy=1.386286 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 3,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500039468
Von Neumann entropy	: 1.3862864795
Eigenvalue min	: 2.494265e-01
Eigenvalue max	: 2.517205e-01
Imaginary part	: < 1e-6 (not shown)

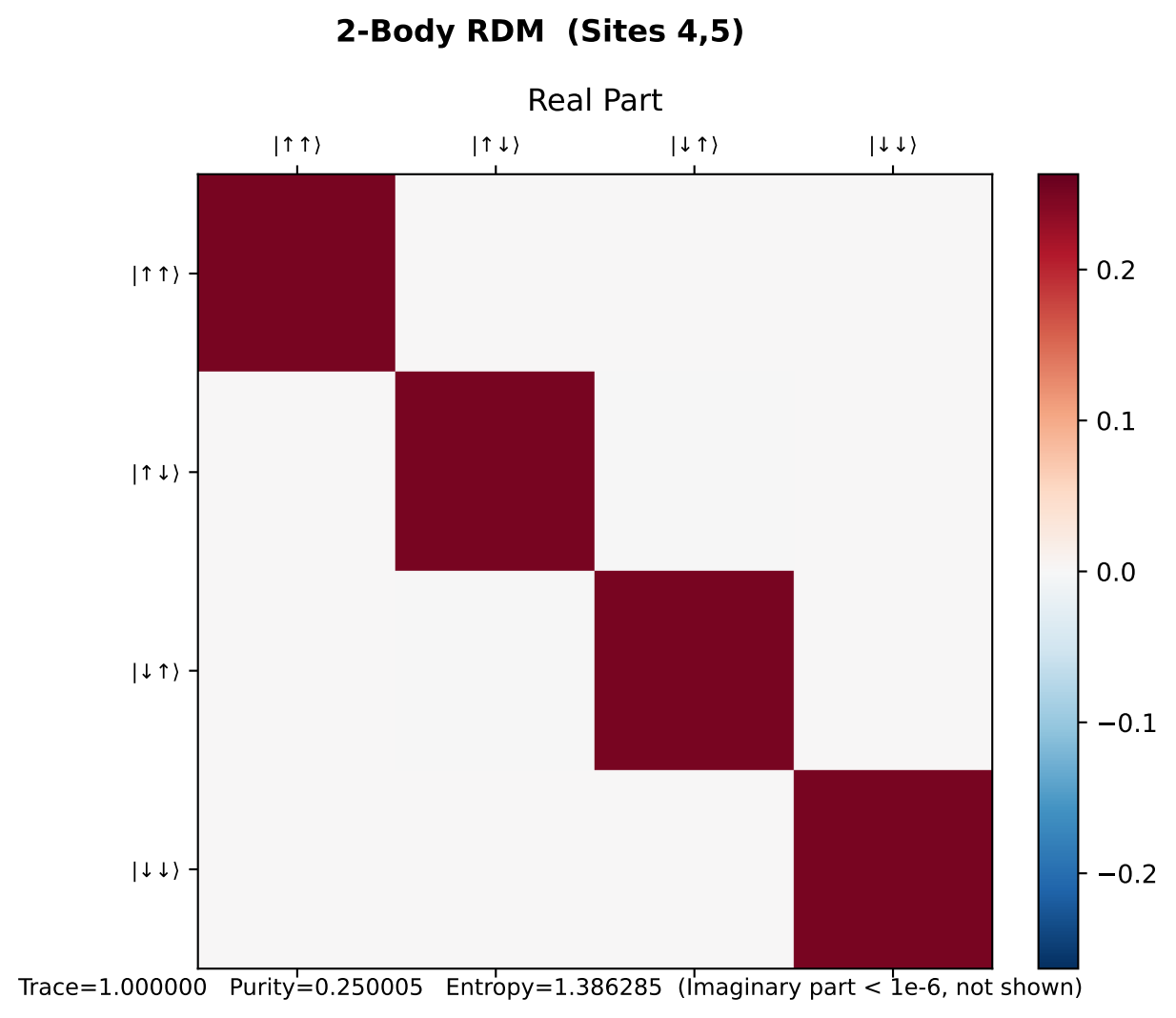
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000395

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 4,5)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250005 Entropy=1.386285 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 4,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500047112
Von Neumann entropy	: 1.3862849544
Eigenvalue min	: 2.493734e-01
Eigenvalue max	: 2.518797e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.25000471

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.